
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived
analysis

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Abstract

1 Tensor decomposition factors a large table of interactions into a small set of low-
2 rank components that can be ranked and inspected. Applying it directly to a
3 conventional robot policy is difficult because softmax, ordinary activations, and
4 normalization interrupt the required algebraic form. We introduce χ -VLA, a
5 compact vision-language-action policy whose learned map uses only bilinear, lin-
6 ear, and rational operations. Its checkpoint therefore defines an explicit rational
7 tensor program, making interactions among image, instruction, state, and action
8 coordinates accessible from the weights. A runtime operator audit of the seed-0
9 20.1M ViT checkpoint finds no incompatible learned operation, and local recon-
10 structions reproduce its deployed rational and bilinear branches near the float32
11 precision floor. Across three independently trained 20.1M ViT policies, closed-loop
12 LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint.
13 A preregistered elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$
14 at the cost of three forward passes. On a separate three-checkpoint convolutional
15 instantiation, independent reconstruction covers all 4,608 joint-attention head-input
16 cases, all 384 joint-FFN module-input cases, and all 384 complete joint-block input
17 cases. The respective worst relative errors are 6.36×10^{-16} , 1.39×10^{-15} , and
18 1.5024×10^{-7} . Exact-attention records naming the same three ViT checkpoint
19 artifacts derive the softmax-free attention decomposition and numerically replay
20 every attention module on one fixed corrected cached input per checkpoint, with
21 worst relative error 2.016119×10^{-7} . In a seed-0 ViT checkpoint, all 36 primary
22 block-head tests favor its rank-128 subspace over equal-rank random controls, with
23 median fidelity ratio 2.44. Separately, prospectively fixed rank-96 visual projectors
24 retain 95.2–98.2% of baseline successes on tasks absent from corrected projector
25 construction, while three equal-rank random projectors retain only 3.3–13.6%.
26 The evidence establishes strong specialist capability and exact layerwise access
27 to trained attention coefficients. It does not establish broad robotic generalization,
28 robust instruction grounding, or an exact end-to-end decomposition of the full
29 policy.

30 1 Introduction

31 Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains
32 millions of learned numbers, but it does not directly reveal which combinations of image regions,
33 instruction tokens, and robot state produce an action. Analysts therefore usually collect activations

34 on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the
35 same structure is present in the weights and whether it causes behavior.

36 Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a
37 decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization
38 finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios
39 of polynomials, its weights define an explicit table of input interactions. Decomposing that table can
40 rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that
41 every learned block preserves this algebraic form. This is an operator-closure claim, not a claim that
42 we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow
43 rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction
44 coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal
45 interactions directly from the checkpoint without first fitting an activation probe. Those factors are
46 candidate structure, not automatically human-readable concepts or causal explanations. The aim is
47 not merely compression. It is a direct path from parameters to candidate mechanisms that can be
48 tested by intervention [3, 4].

49 Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark
50 success while ignoring the instruction, and small action errors compound in closed loop. Standard
51 softmax, nonlinear activations, and square-root normalization break the explicit polynomial form.
52 Continuous actions also cannot rely on the external categorical sampling used by a language model.
53 A practical design must recover these functions without losing precise control.

54 We study the narrower but testable question:

55 *Can a specialist VLA retain strong closed-loop capability when every deployed*
56 *learned operation is polynomial or rational, while supporting exact weight analy-*
57 *sis?*

58 We test capability and structural analysis in two fully rational encoder instantiations. The same three
59 ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate
60 convolutional instantiation tests whether the structural certificates transfer across visual encoders.

61 Our contributions are:

- 62 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
63 supplies input-specific scaling without leaving the rational function class.
- 64 2. We evaluate three fully rational 20.1M ViT policies on 1,500 canonical LIBERO-Object
65 trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches
66 $467/500 = 93.4\%$.
- 67 3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed
68 rational and bilinear branches from their weights.
- 69 4. We extend exact weight-derived decomposition through individual attention layers with a
70 reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate
71 closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes,
72 versus 3.3–13.6% for random projectors.

73 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
74 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
75 policy makes exact weight structure directly accessible.

76 **2 A rational tensor robot policy**

77 **2.1 Tensor-compatible primitives**

78 Each primitive below replaces a familiar transformer component while keeping an explicit coeffi-
79 cient description. Linear maps contribute first-order terms. Multiplying learned linear projections
80 creates higher-order interactions whose coefficients remain functions of the weights, rather than an
81 unstructured nonlinear black box.

82 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 83 layer to represent bias, linear, and quadratic terms in one contraction.

84 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

85 The inner sum is an order-three table indexed by output coordinate and two input coordinates. The
 86 three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization,
 87 without materializing every table entry.

88 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

89 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value,
 90 and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike
 91 softmax attention, every multiplicative interaction can be expanded into coefficients determined by
 92 the trained weights.

93 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 94 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
 95 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

96 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
 97 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
 98 operations update this pair. The model therefore keeps input-specific rescaling while remaining
 99 exactly representable as a rational tensor network. Training and deployment use r itself. The
 100 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
 101 not uniform agreement with RMSNorm. Appendix A.3 certifies pole-free denominators and deployed
 102 arithmetic fidelity while retaining that boundary.

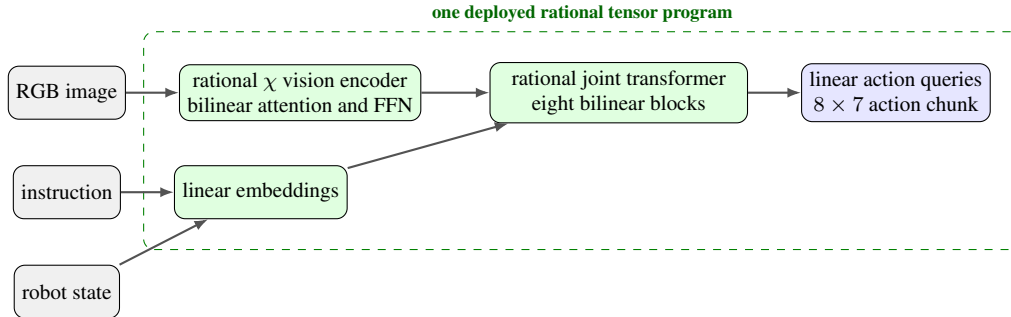


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

103 2.2 Architecture and training

104 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
 105 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
 106 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both
 107 share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-
 108 query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision

blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional files receive the joint-attention and rational-normalizer certificates reported in Appendix A.3.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M ViT checkpoint at runtime and reconstruct representative operations from saved weights.

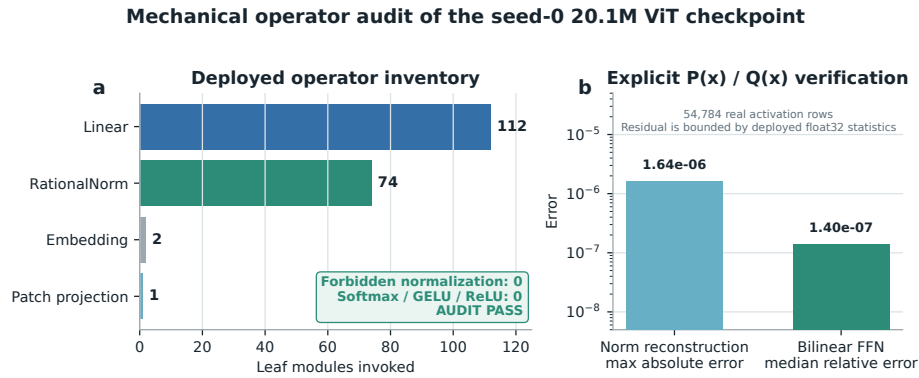


Figure 2: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 50 canonical trials per task
- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

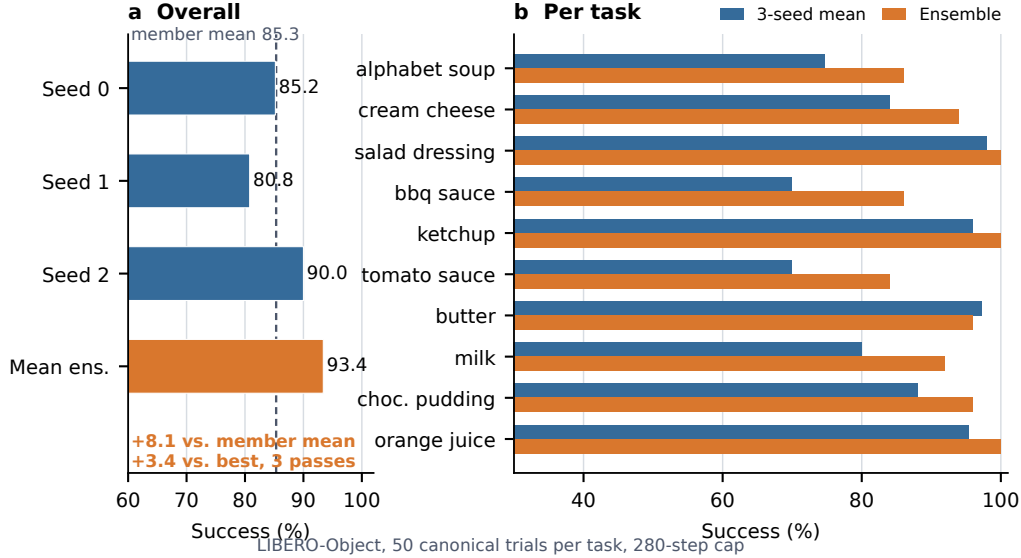


Figure 3: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ -VLA, rational ViT	20.1M	1 pass	85.3% $\pm$ 4.6%
χ -VLA, prediction-mean ensemble	$3 \times 20.1M$	3 passes	93.4%

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and

value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . On one fixed corrected cached input per checkpoint, we numerically replay all four vision and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5} . This residual comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

We separately certify three convolutional checkpoints from the same rational architecture family. Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs per checkpoint yield 4,608 head-input cases. Every output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , below the frozen 10^{-6} gate. On the same inputs, independent CP-factor evaluation covers all 384 joint-FFN module-input cases at worst relative error 1.3871×10^{-15} against a 10^{-10} gate. Independent complete-block equations cover all 384 block-input cases at worst relative error 1.5024×10^{-7} against a 10^{-4} gate. The complete-block comparison crosses a deployed PyTorch float32 block call and a NumPy float64 reconstruction. It is input-conditioned and per-block, not a sequential end-to-end reconstruction of the joint transformer or policy.

A separately frozen ViT audit partitions joint attention into vision, instruction, robot-state, and action-query source tokens. Across 128 deterministic, official-task-balanced inputs for each of the three capability checkpoints, all 36,864 head-input cases and 147,456 source-group evaluations are finite, with worst reconstruction error 5.0741×10^{-7} against a 10^{-6} gate. Across the 3,072 projected module-input cases, mean coherent-energy shares are 67.7% vision, 19.2% robot state, 7.9% action query, and 5.1% instruction. These input-conditioned magnitudes are descriptive rather than causal importance. Familiar-prompt permutations are also descriptive and do not establish grounding or closed-loop success.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14} .

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

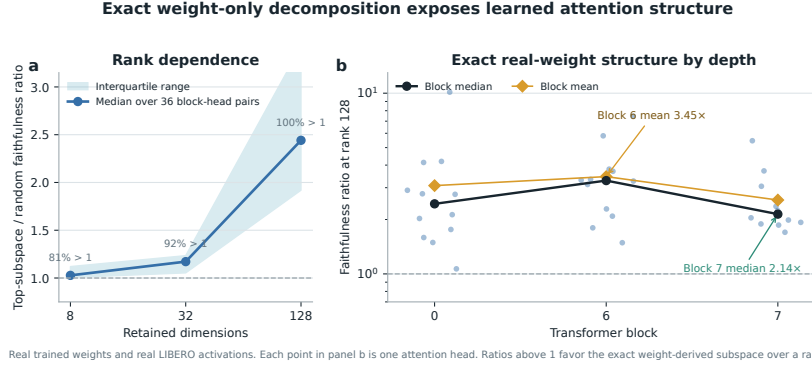


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

A separate intervention tests behavioral sufficiency at the visual-to-joint interface rather than deriving a projector from the exact attention factors. An action-Jacobian projector estimated on official Object tasks 0–3, with rank 96 fixed before corrected outcomes, retains 259/272 full-policy successes on tasks 8–9 across three ViT checkpoints. These tasks are absent from corrected projector construction. An activation-energy projector retains 267/272, while three equal-rank random projectors retain 37/272, 9/272, and 22/272. All ten tasks appeared during policy training. This establishes a structured low-dimensional visual bottleneck relative to random projectors, not unseen-task generalization, unique action-Jacobian specificity, or a causal link to the exact weight factors. Appendix A.2 reports the paired controls and stability intervals.

5 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight auditing in a strong continuous vision-language policy.

228 **VLA robustness.** Recent evaluations show that standard LIBERO success can coexist with weak
229 robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist
230 capability setting rather than evidence of broad generalization.

231 6 Limitations and conclusion

232 χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop
233 capability. The capability and exact-attention records name the same three ViT checkpoint artifacts.
234 The deployed seed-0 operator path also passes a mechanical audit.

235 The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions,
236 so these results do not establish broad robotic generalization or robust instruction grounding. Exact
237 reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of
238 the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability
239 result.

240 Within that scope, polynomial and rational restrictions are compatible with strong closed-loop
241 control and provide a precise weight-derived analysis surface. The ViT records connect capability,
242 exact attention, and the reduced-Gram screen through common checkpoint basenames. Separate
243 convolutional files replicate the joint-attention equation certificate across another visual encoder. We
244 do not claim a direct coefficient-edit interface.

245 **Reproducibility.** Training and evaluation use fixed task lists, explicit camera and action conventions,
246 canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical
247 Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline
248 pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A
249 source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10
250 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized
251 images matched exactly for every frame. The anonymous artifact releases raw structural-certificate
252 JSONs and all 2,000 episode rows behind the ViT member and ensemble results. Immutable
253 SHA-256 manifests and standard-library verifiers independently recompute the capability totals,
254 visual-bottleneck counts and intervals, and the attention, FFN, complete-block, modality-partition,
255 and RationalNorm certificate decisions. Preflight checks fail if canonical initial states cannot be
256 loaded.

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A Supporting results and decisions

A.1 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.2 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

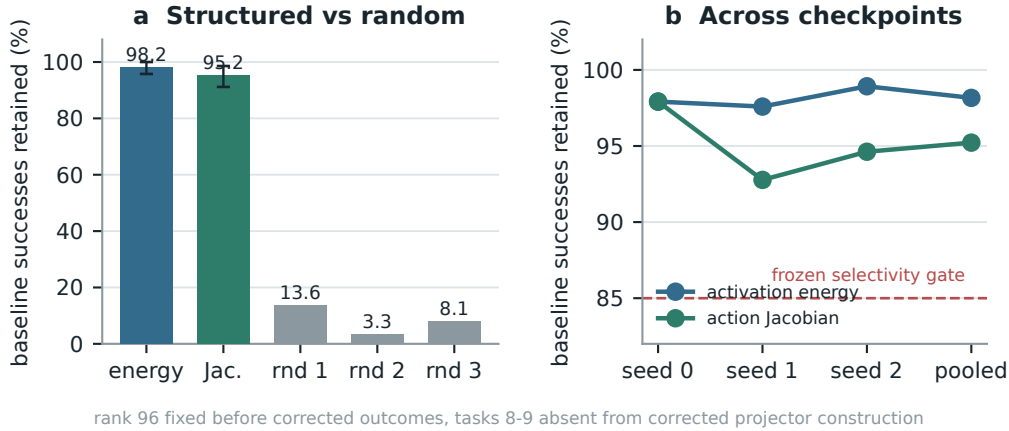


Figure 5: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This intervention is data-derived and remains separate from the exact weight-only attention decomposition.

301 A.3 Numerical scope of rational conversion

302 The deployed degree-[2/2] least-squares rational fit is exact as a rational computation even though it
 303 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 304 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 305 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 306 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 307 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016
 308 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass
 309 that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric,
 310 defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a
 311 frozen 10^{-3} gate. This primary certificate concerns pole-free denominators and floating-point fidelity
 312 to the deployed rational operator. It does not claim uniform agreement with exact RMS scaling or
 313 measure matched alternative-normalizer runtime overhead.

314 The joint reconstruction ladder therefore extends from individual attention heads through FFNs to
 315 complete joint blocks on the same fixed inputs. The full-block certificate remains per-block and does
 316 not sequentially reconstruct the full transformer. The ViT source-partition audit is likewise layerwise
 317 and input-conditioned. Figure 6 reports the frozen numerical gates and the descriptive depth profile
 318 without assigning causal importance.

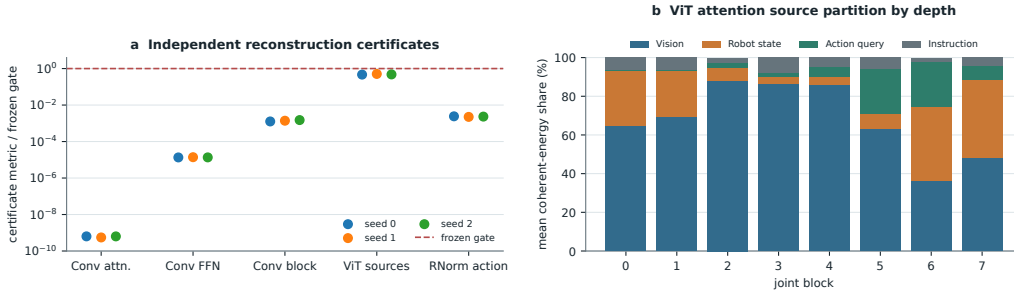


Figure 6: **Frozen reconstruction ladder and descriptive ViT source partition.** Panel (a) reports observed-to-gate ratios for independent Conv attention, joint-FFN, and complete joint-block reconstruction, plus the ViT source-partition identity and primary RationalNorm action fidelity. All plotted primary numerical gates pass. Panel (b) shows descriptive mean coherent-energy shares after the attention output projection across 384 checkpoint-input cases per joint block. The depth trend is not a causal importance or grounding result. Complete-block replay is per-block rather than sequential end-to-end policy reconstruction.